

Data Quality Report

messy_survey.csv · generated 15 July 2026, 20:41 UTC

	Before	After
Quality score	74/100	87/100
Rows	612	600
Columns	18	19
Duplicate rows	12	0

Executive summary

The dataset arrived with 612 rows across 18 columns and scored 74/100 on the quality index. After 13 recorded operations it scores 87/100, a change of +13 points.

Treat these columns as skip-logic gated and do not impute them: Q9_followup. Analyse them only among respondents who were asked.

Potentially identifying columns were detected: respondent_id, respondent_name, email, free_text_comment. De-identify before sharing this dataset outside the study team.

Every change is listed in the audit trail below and is reproducible via the exported cleaning script.

Summary generated from the computed figures above (no model was available).

What we found

Column	Issue	Conf.	Detail
	duplicate rows	95%	12 exactly duplicated rows (2.0%). Usually a double submission or a merge run twice. Check it isn't legitimate repeat measurement before removing.
gender	sentinel missing	95%	85 values (13.9%) look like missing-data codes: ['Prefer not to say']
Q1	sentinel missing	95%	94 values (15.4%) look like missing-data codes: ['Prefer not to say']
Q2	likert scale	95%	Matches the 'frequency_5' scale. Currently unordered — median and sort will be wrong until it's made ordinal.
Q9_followup	likert scale	95%	Matches the 'agreement_5' scale. Currently unordered — median and sort will be wrong until it's made ordinal.
monthly_income	sentinel missing	95%	13 values (2.1%) look like missing-data codes: [-99.0]
survey_date	date like	95%	100% of values parse as dates but the column is text, in 4 different formats. As text it sorts alphabetically, so any trend over time is wrong.
gender	case variants	92%	7 categories collapse to 5 once case and whitespace are normalised (167 rows affected). e.g. ['Female ', 'female']
state	case variants	92%	9 categories collapse to 6 once case and whitespace are normalised (203 rows affected). e.g. ['Lagos', 'LAGOS', 'Lagos']
Q7	multiselect group	90%	5 columns are one multi-select question. Blank here means 'not ticked', not 'missing' — filling them would invent data.

Column	Issue	Conf.	Detail
consent	constant columns	90%	1 column(s) hold the same value in every row: ['consent']. They carry no information. Check they aren't the residue of an upstream filter before dropping.
age	sentinel missing	90%	38 values (6.2%) look like missing-data codes: [999]
free_text_comment	sentinel missing	90%	191 values (31.2%) look like missing-data codes: ['- ', "Don't know"]
Q1	likert scale	83%	Matches the 'agreement_5' scale. Currently unordered — median and sort will be wrong until it's made ordinal.
respondent_id	pii	80%	Column name suggests name. De-identify before sharing or publishing.
respondent_name	pii	80%	Column name suggests name. De-identify before sharing or publishing.
email	pii	80%	Column name suggests contact. De-identify before sharing or publishing.
age	implausible range	75%	38 value(s) fall outside 0–120, the plausible range for age in years. Often a leftover missing-data code. Flagged, not changed — check them before deciding.
Q9_followup	structural missing	55%	68.5% blank — possibly skip logic, possibly poor response. Check the questionnaire for a gating question. Do NOT impute; analyse only among those who were actually asked.
free_text_comment	pii	50%	Column name suggests free text. De-identify before sharing or publishing.

Audit trail

Every change made to the data, in order. This report is reproducible: the exported cleaning script regenerates the cleaned dataset from the raw file using only pandas.

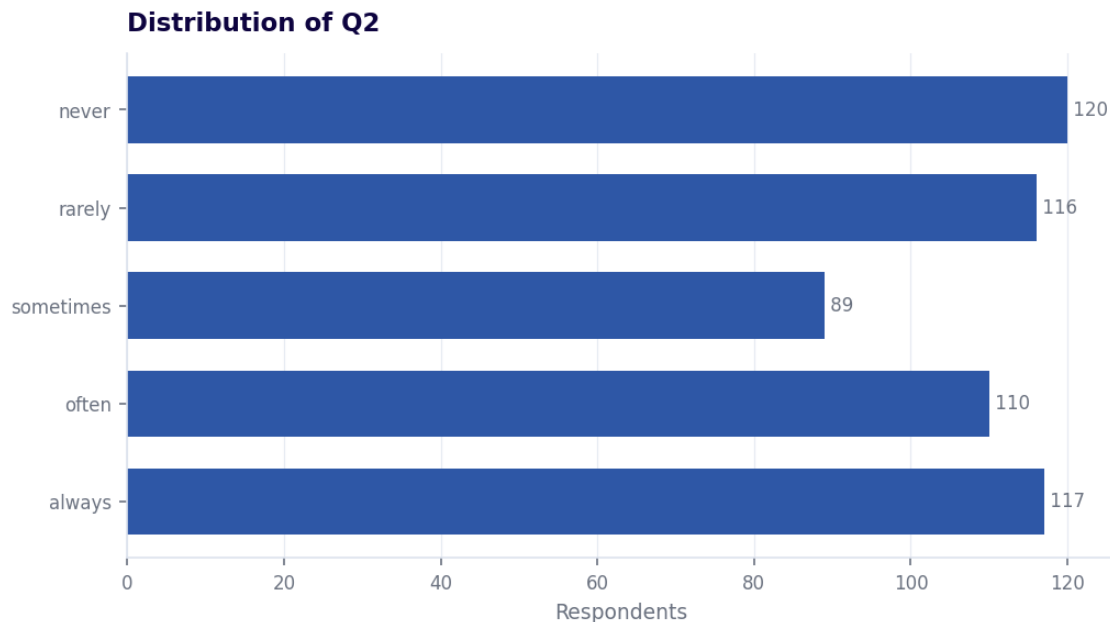
#	Operation	Rows	Shape	Why
1	collapse_multiselect stem=Q7, members=['Q7_Part_1', 'Q7_Part_2', 'Q7_Part_3', ...]	612	612x18 -> 612x20	5 columns are one multi-select question. Blank here means 'not ticked', not 'missing' — filling them would invent data.
2	drop_constant_columns columns=['consent']	612	612x20 -> 612x19	1 column(s) hold the same value in every row: ['consent']. They carry no information. Check they aren't the residue of an upstream filter before dropping.
3	recode_sentinel_missing column=gender, values=['Prefer not to say']	85	612x19 -> 612x19	85 values (13.9%) look like missing-data codes: ['Prefer not to say']
4	recode_sentinel_missing column=Q1, values=['Prefer not to say']	94	612x19 -> 612x19	94 values (15.4%) look like missing-data codes: ['Prefer not to say']
5	recode_sentinel_missing column=monthly_income, values=[-99.0]	13	612x19 -> 612x19	13 values (2.1%) look like missing-data codes: [-99.0]
6	recode_sentinel_missing column=age, values=[999]	38	612x19 -> 612x19	38 values (6.2%) look like missing-data codes: [999]
7	recode_sentinel_missing column=free_text_comment, values=["Don't know", '-']	191	612x19 -> 612x19	191 values (31.2%) look like missing-data codes: ['- ', "Don't know"]
8	coerce_datetime column=survey_date	612	612x19 -> 612x19	100% of values parse as dates but the column is text, in 4 different formats. As text it sorts alphabetically, so any trend over time is wrong.
9	order_likert column=Q2, order=['never', 'rarely', 'sometimes', 'often'...]	563	612x19 -> 612x19	Matches the 'frequency_5' scale. Currently unordered — median and sort will be wrong until it's made ordinal.
10	order_likert column=Q9_followup, order=['strongly disagree', 'disagree', 'neutr...]	193	612x19 -> 612x19	Matches the 'agreement_5' scale. Currently unordered — median and sort will be wrong until it's made ordinal.
11	map_categories column=gender, mapping={'female': ' Female ', 'Male': 'male'}	167	612x19 -> 612x19	7 categories collapse to 5 once case and whitespace are normalised (167 rows affected). e.g. [' Female ', 'female']
12	map_categories column=state, mapping={'LAGOS': ' Lagos', 'Lagos': ' Lagos',...}	203	612x19 -> 612x19	9 categories collapse to 6 once case and whitespace are normalised (203 rows affected). e.g. [' Lagos', 'LAGOS', 'Lagos']
13	drop_duplicates	12	612x19 -> 600x19	12 exactly duplicated rows (2.0%). Usually a double submission or a merge run twice. Check it isn't legitimate repeat measurement before removing.

Charts

Drawn from the cleaned data above, not the raw file. Each one notes what to look for — and what would make it misleading.

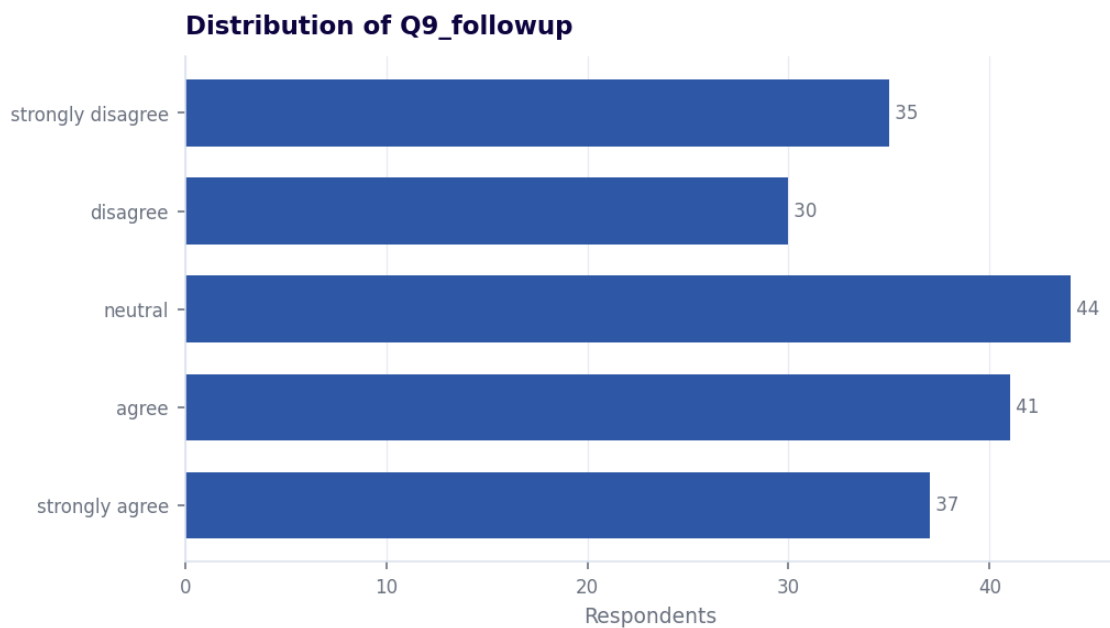
Distribution of Q2

Q2 is an ordered scale, so the bars come out in scale order rather than alphabetically. Look for the shape: a flat spread means genuine disagreement, a pile-up at one end often means the question was leading.



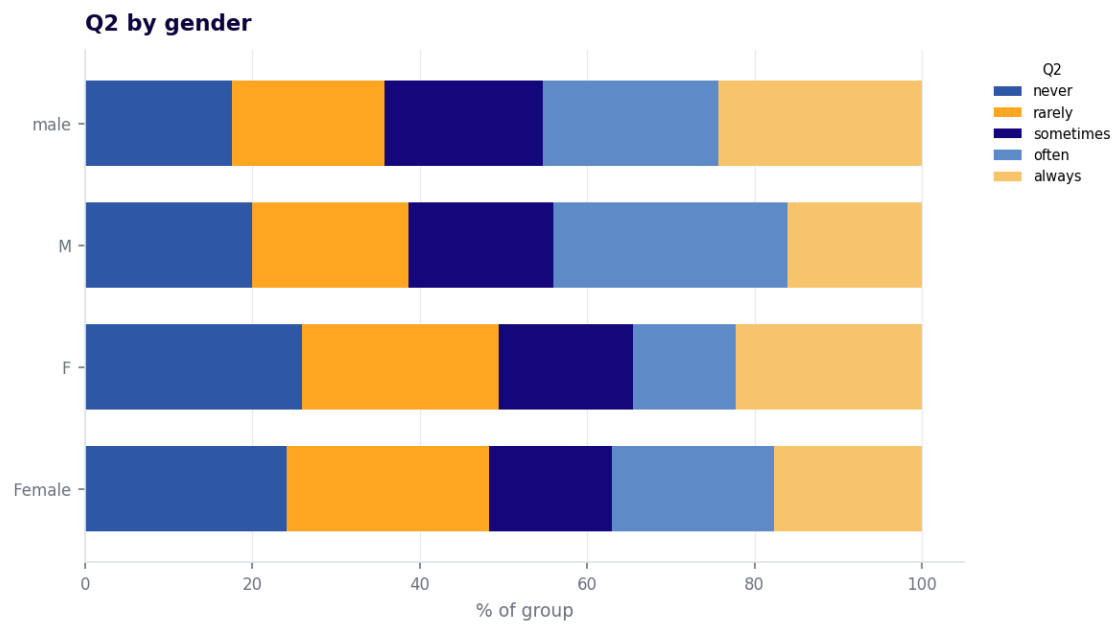
Distribution of Q9_followup

Q9_followup is an ordered scale, so the bars come out in scale order rather than alphabetically. Look for the shape: a flat spread means genuine disagreement, a pile-up at one end often means the question was leading.



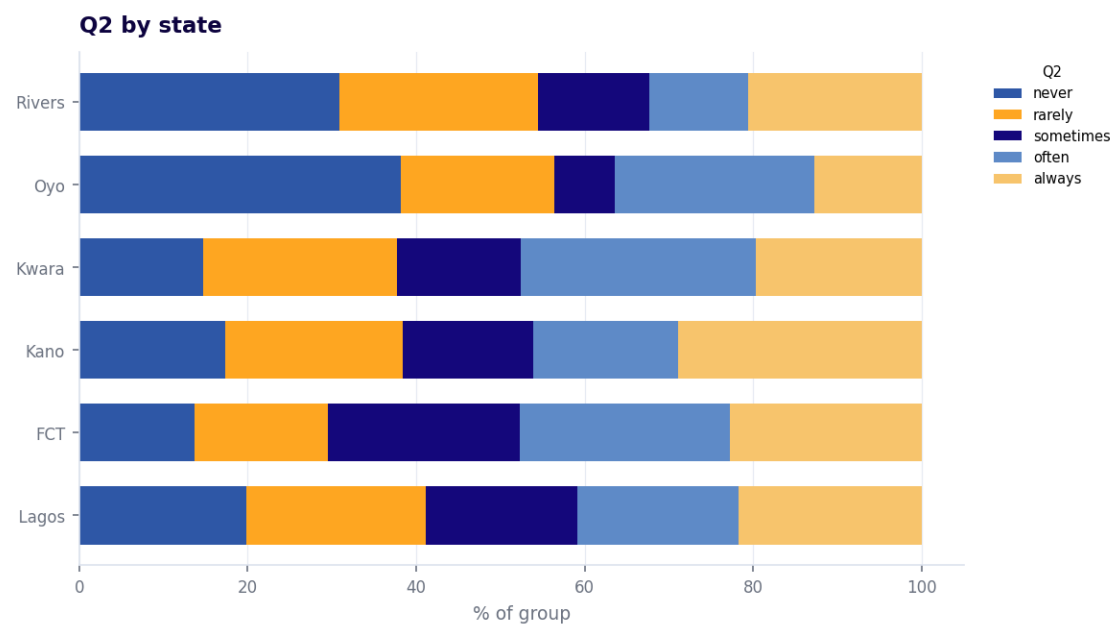
Q2 by gender

Splits Q2 by gender as row percentages, so groups of different sizes stay comparable. This is usually the finding — a difference here is what the survey was for.



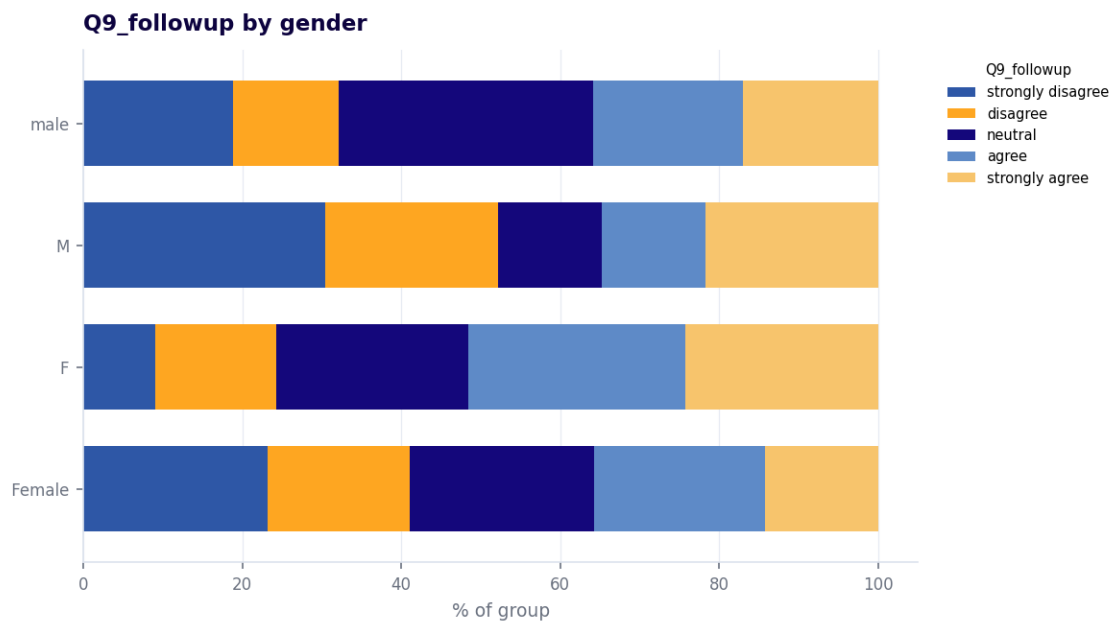
Q2 by state

Splits Q2 by state as row percentages, so groups of different sizes stay comparable. This is usually the finding — a difference here is what the survey was for.



Q9_followup by gender

Splits Q9_followup by gender as row percentages, so groups of different sizes stay comparable. This is usually the finding — a difference here is what the survey was for.



Q9_followup by state

Splits Q9_followup by state as row percentages, so groups of different sizes stay comparable. This is usually the finding — a difference here is what the survey was for.

